IVT, EVT

Theorem

Intermediate Value Theorem (IVT)

If f is continuous on the interval $[a, b]$, and either $f(a) < \alpha < f(b)$ or $f(a) > \alpha > f(b)$, then there exists a point $c \in (a, b)$ such that $f(c) = \alpha$.

Proposition

The Bisection Method

<u>Step 1</u> - Find 2 points a_0 and b_0 $(a_0 < b_0)$ such that $F(a_0)$ and $F(b_0)$ have different signs.

IVT guarantees a point c between a_0 and b_0 such that $F(c) = 0$.

<u>Step 2</u> - Find the midpoint $d = \frac{a_0 + b_0}{2}$. Evaluate $F(d)$

<u>Step 3</u> - If $F(a_0)$ and $F(d)$ have the same sign, let $a_1 = d$ and $b_1 = b_0$. Otherwise $a_1 = a_0$ and $b_1 = d$. The new interval $[a_1, b_1]$ contains a solution that has a size of $\frac{1}{2}(b_0 - a_0)$.

<u>Step 4</u> - Repeat Steps 2 and 3 to get new intervals $[a_2, b_2]$, $[a_3, b_3]$, .., and the kth interval will have $\frac{1}{2^k}(a_0 - b_0)$ as its length.

<u>Step 5</u> - Stop when $\frac{1}{2^{k+1}}(b_0 - a_0) < \varepsilon$.

Let $d = \frac{a_k + b_k}{2}$, then the maximum distance from d to a solution c is less than ε $(|d - c| < \varepsilon)$.

Def'n

Suppose $f: I \to \mathbb{R}$ where I is an interval.

- c is a <u>global maximum</u> for f on I if $c \in I$ and $f(x) \leq f(c)$ for all $x \in I$
- c is a <u>global minimum</u> for f on I if $c \in I$ and $f(x) \geq f(c)$ for all $x \in I$.
- c is a <u>global extremum</u> if it is either a global maximum or global minimum

Theorem

Extreme Value Theorem (EVT)

Suppose f is continuous on $[a, b]$. Then there exists two numbers, $c_1, c_2 \in [a, b]$ such that $f(c_1) \leq f(x) \leq f(c_2)$ for all $x \in [a, b]$.

PRACTICE PROBLEMS

1. Let $d_1(t)$ represent the distance from UW on Friday
Let $d_2(t)$ represent the distance from UW on Monday
Note that in both cases $t = 0$ starting at 10 AM.

Let $m > 0$ represent the distance from UW to Montreal.

$$q(t) = d_1(t) - d_2(t)$$
$$q(0) = d_1(0) - d_2(0) = 0 - m = -m$$
$$q(8) = d_1(8) - d_2(8) = m - 0 = m$$

Since $q(0) < 0$ and $q(8) > 0$, by IVT there exists some number c with $0 < c < 8$ such that $q(c) = 0$.

$$q(c) = 0 = d_1(c) - d_2(c) \Rightarrow d_1(c) = d_2(c)$$

2. (a) Note that $p(x)$ is a polynomial, which is continuous everywhere. Since $p(0) < 0$ and $p(1) > 1$, by IVT, there exists a point $c \in (0,1)$ so that $p(c) = 0$. Thus, p has a root between $x = 0$ and $x = 1$.

(b) Since $p(\frac{1}{2}) < 0$ and $p(1) > 0$, by IVT, there exists a point $c \in (\frac{1}{2}, 1)$ such that $p(c) = 0$. Observe that $(\frac{1}{2}, 1)$ is an interval of length $\frac{1}{2}$ which contains a root of $p(x)$.

3. (a) Since f is continuous and on a closed interval, EVT guarantees two values c_1 and c_2 such that $f(c_1) \leq f(x) \leq f(c_2)$. $c_1 = 2$, $c_2 = 0$.

(b) No, f is continuous but not on a closed interval.

(c) Yes, EVT applies. $c_1 = 4$, $c_2 = 4$.

(d) No, f is not continuous on $\frac{\pi}{2}$.

(e) No, f is not continuous at 0.

4. (a) T

(b) T

(c) F

(d) F

5.

$$v(t_0) = \lim_{h \to 0} \frac{v(t+h) - v(t)}{h}$$

$$= \lim_{h \to 0} \frac{-9.8(1+h)^2 + 30(1+h)^2 + 12 + 9.8 - 30 - 12}{h}$$

$$= \lim_{h \to 0} \frac{10.4h - 9.8h^2}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{h}(10.4 - 9.8h)}{\cancel{h}}$$

$$= 10.4$$

6. (a)

$$f'(2) = \lim_{h \to 0} \frac{f(2+h)^4 - 2^4}{h}$$

$$= \lim_{h \to 0}$$

$$= \lim_{h \to 0}$$